

Chapter 19 - The Growth of Functions

3 pts

1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

(a) ☐ True ☐ False $11n^{11} + 6n + 2$ is $O(n^{11})$

(b) ☐ True ☐ False $n \log n + 2^n + n$ is $O(n \log n)$.

(c) ☐ True ☐ False If $f_1(n)$ is $O(g_1(n))$ and $f_2(n)$ is $O(g_2(n))$, then $f_1(n)f_2(n)$ is $O(g_1(n)g_2(n))$.

Chapter 20 - The Integers

1 pts

1. Prove that if $a > 0$ and $b < 0$, then $ab < 0$.

1 pts

2. Prove that if $m^2 = n^2$, then $m = n$ or $m = -n$.
(Hint from algebra: $a^2 - b^2 = (a + b)(a - b)$.)

1 pts

3. Prove that if n is an integer, then $n^2 \geq 0$.

Chapter 21 - The **divides** Relation and Primes

1 pts

1. For positive integers, a and b , if the quotient when a is divided by b is q , what are the possible quotients when $a + 1$ is divided by b ?

1 pts

2. Prove or give a counterexample: If p is a prime, then $2p + 1$ is a prime.

1 pts

3. Suppose the remainder when a is divided by b is r . Determine the remainder when $a + 2b$ is divided by b . More generally, if k is any integer, determine the remainder when $a + kb$ is divided by b .

1 pts

4. Show that for any integer n , there are n consecutive non-prime integers.